



Ramakrishna Mission Vivekananda Educational & Research Institute

Belur Math, Howrah, West Bengal
Department of Computer Science

MSc Big Data Analytics, I Semester, Midterm Examination

Date: 18 Sep 2024

Course : CS110: Data Structures and Algorithms

Time: 2 hrs

Instructor : Punyeshwarananda

Max marks: 60

Student Id:

Student Signature: _____

Answers should be clear, concise, and complete to earn the full allotted marks. All steps in the problem-solving process should be clearly stated.

Answer all the questions

1. Prove or disprove the following: i). $\log_b n$ is $O(n^d)$ for all $b > 0, d > 0$ ii). $n!$ is $2^{\Theta(n \log n)}$ (5+5)
2. (a) Give the Python code for solving tower of Hanoi puzzle for n disks. Derive its time complexity. (6+4)
(b) Show that if $d(n)$ is $O(f(n))$ and $e(n)$ is $O(g(n))$, then the product $d(n)e(n)$ is $O(f(n)g(n))$.
3. (a) Describe the divide-and-conquer design pattern. Derive the time complexity of merge-sort algorithm. (6+4)
(b) Give the Python code for finding the n^{th} element in a Fibonacci series in $O(n)$ time complexity (linear time) using recursion.
4. (a) What is the time complexity of the code shown in listing(a) below? (2+5+3)
(b) What is the output of the code shown in listing(b)?
(c) Arrange the following functions in the increasing order of growth:
 $(\sqrt{2})^{\log n}, e^n, 4^{\log n}, 2^n, n!, (\log n)!$
5. (a) What is the time complexity of the following recurrence equations: (4+4+2)
i). $T(n) = 3T(\frac{n}{3}) + \frac{n}{2}$
ii).
$$T(n) = \begin{cases} 2T(n-1) - 1 & \text{if } n > 0 \\ 1 & \text{otherwise} \end{cases}$$

(b) The Master's theorem fails for this recurrence $T(n) = T(\frac{n}{2}) + n(2 - \cos n)$. Explain why?
6. (a) Derive the amortized bound for performing a series of N-append operations on an initially empty dynamic array implemented using fixed size increment strategy. (5+5)
(b) Write a Python class called NonPrimes for listing all non-prime numbers up to n (inclusive)
The end user should be able to call your class in this manner:

```
pobj = NonPrimes(19) # instantiation of your implemented class NonPrimes
pobj.listall() # should print all non-prime numbers ie. 1,4,6... 18
```

Code Listing Referenced in the Questions

listing (a)

```
def func(n):
    c, i = 0, 1
    while i*i <= n:
        c += 1
        i += 1
    return c
```

listing (b)

```
lst = [21,42,7,11,30,44,23,0]
print(lst[:1:-1])
print(lst[:-2:3])
lstA = [i for i in lst if ((i & 0x1) or (not i))]
print(lstA)
```